Young Innovators Programme (YIP) – Kerala

Kerala Development and Innovation Strategic Council (K-DISC)

Complete Programme Guide & Thematic Problem Compendium

YIP 1.0 (2018) to YIP 9.0 (2026) — 27 Themes — 1972 Problem Shelf (login-gated)

Official portal	https://yip.kerala.gov.in/
Themes page	https://yip.kerala.gov.in/theme
About	https://yip.kerala.gov.in/about
Problem submission	https://yip.kerala.gov.in/problemstatement
Student signup	https://yip.kerala.gov.in/student/signup
K-DISC	https://kdisc.kerala.gov.in/en/young-innovators-programme-yip

Scope note (important): This file contains **all public information** obtainable without login: programme overview, governance, eligibility, journey, YIP 1.0–8.0 statistics, all **27 official themes** with focus areas and representative challenge directions, notable past projects, benefits, timelines, and exact steps to open the full 1972-item problem shelf from inside a student account. The site serves the problem shelf only after email-verified login + institution selection (`/student/*`, Next.js + auth), so no anonymous crawler (UmbraCrawl included) can list the 1972 verbatim items. UmbraCrawl was attempted (crawled 9+17 pages, 10+ blocked by client-render/auth wall) and direct fetch of `/student` returns 404 without session. Do not treat the representative prompts in §5 as verbatim shelf text – use them to shortlist a theme, then copy the verbatim text from your dashboard.

Contents

Programme at a glance

Young Innovators Programme (YIP), launched 2018 by the Government of Kerala and conceptualised by K-DISC, equips students aged 13–37 with 21st-century skills (critical thinking, creativity, collaboration, communication) via Design Thinking, Root Cause Analysis, and DUI (Doing, Using, Interacting) innovation. Teams of 2–5 address curated real-world second-generation Kerala problems using IR 4.0 technologies, under a quadruple-helix model (academia + industry/startups + government + civil society).

- **Motto:** Igniting the Innovative Minds of Kerala.
- **Streams:** (1) School (Classes 8–12), (2) Technical / collegiate, (3) Medical and allied, (4) Employment / training / professionals.
- **Flow:** Ideator registration → idea submission (across 30 areas / 27 listed themes *or* from 1972-item problem shelf) → L0 evaluation → L1 (preliminary) evaluation → training / mentoring → maker camp → protothon → district / state winners → grants, incubation, patents, startups, exit.
- **Scheme:** Approved by Council of Ministers 30-12-2022; 50-page scheme PDF via DTE Kerala.
- **2025-26 highlights (K-DISC Annual Report):** YIP 8.0 – 12.57 lakh participants across 8,658 institutions, 47,641 ideas, 1,026 state-level winners; indigenous YIP portal built for end-to-end management; Rs. 4 crore total winner grants disbursed.

Governance

Steering Committee includes Directors of Technical, Collegiate, Medical, General Education, Ayurveda, Homeopathy, Principal Secretaries (Higher Ed, LSGD, General Ed, Labour & Skills), Health, Employment & Training, SC/ST/Women-Child/Social Justice/SSK, CEOs of KSUM and KITE, K-DISC EVC, Member Secretary, Executive Director (PCDIS), Programme Head YIP. Advisory Committee comprises Vice-Chancellors of Kannur, KVASU, Calicut, KAU, KUHS, Sanskrit, KUFOS, CUSAT, NUALS, MG, KTU, Digital, Malayalam, Kerala, Central Kerala, Kalamandalam universities plus K-DISC leadership.

Key numbers (official homepage, 2026)

Edition	Registered	Ideas	District winners	State winners
1.0 (2018)	1,320	204	44	22
2.0 (2019)	3,950	1,003	210	102
3.0 (2020)	13,002	3,747	234 teams	96 teams
4.0 (2021)	1,02,512	9,404	229 teams	114 teams
5.0 (2022)	1,58,965	17,877	508 teams	98 teams
6.0 (2023)	2,13,473	25,352	274 teams	45 teams
7.0 (2024)	2,77,248	26,320	318 teams	46 teams
8.0 (2025)	4,82,738	47,641	437 teams	74 teams

Impact: female participation ∼53.3% registration / 50.3% state winners; grants Rs. 25,000 per district team, Rs. 50,000 per state team; prototypes, patents filed/under process, startups (Blukaze, AgenticNex AI, Kalpa Lights, Coearth, Redfox Robotics, URAV, Cashcrow, etc. – full list on https://yip.kerala.gov.in/).

Who can apply and how

1. Open https://yip.kerala.gov.in/student/signup. Fields: Full Name, Phone, Gender, Email (verify via OTP/link first), District (dropdown loads districts), Institution (approved-institution dropdown loads after district), Password + Confirm.
2. Verify email before "Create account" enables. Then Sign in at https://yip.kerala.gov.in/login.
3. Form team of 2–5. Choose either (a) open idea under any of 27 themes, or (b) pick a shelf problem (1972 items filterable by theme/district/department) from inside dashboard → submit idea canvas.
4. YIP 8.0 window for college categories was 01-Oct-2025 to 15-Nov-2025 (DTE circular DTETVM/8553/2025). YIP 9.0 launched 2026 – check dashboard / Instagram https://www.instagram.com/yipkerala/ for current window.
5. Each institution should nominate one faculty YIP Facilitator and support K-DISC district representatives for campaigns.

Benefits, stages, timeline

Training in design thinking and innovation leadership, specialised mentorship in new-product Fuzzy Front End, maker camp + protothon (Protothon 1.0: 60 teams, 10 funded), activity points, certificates, grace marks for Govt. school state winners, patent support, grant funding, incubation via KSUM, career foundation through real-world problem solving. Timeline per cohort spans roughly one year; exact YIP 9.0 dates appear only inside login and on socials.

Notable past projects (official /about)

Lipid liquid-crystal fish-adulteration detector; anti-bedsore clothing; advanced fumigation device; water-hyacinth value products; marijuana detector; pneumatic exoskeleton; IoT concrete-maturity meter; coconut-oil bio-resin; semi-auto rubber mixer; auto solar-panel cleaner; unified pet-health app; bio-bin; energy-saving vessel; multi-purpose agri vehicle; aero water maker.

The 27 official themes – full compendium

Below are the **verbatim 27 theme titles** from https://yip.kerala.gov.in/theme. Under each, *Focus* summarises the domain and *Representative challenge directions* gives typical shelf-style prompts to guide ideation. These prompts are **not** verbatim shelf copies – always replace with the exact shelf text from your dashboard before submission (shelf requires ≥100-word problem framing: challenge, who is affected, gaps, desired outcome).

1. Agriculture, Plant Science & Smart Farming

Focus: paddy/plantation productivity, soil health, precision irrigation, pest/disease, farm mechanisation for smallholders. *Directions:* low-cost soil-moisture + fertigation advisory for laterite soils; drone/image pest detection for banana/pepper; small-farm harvester for fragmented plots.

2. Food Technology & Agro Processing

Focus: post-harvest loss, value addition (jackfruit, banana, fish, spices), cold chain, food safety. *Directions:* solar micro-dehydrator for SHGs; low-cost fish chill-box for vendors; rapid adulteration kit for milk/chilli.

3. Dairy, Animal Husbandry & Poultry Sciences

Focus: milk yield, mastitis, feed cost, poultry heat stress, vet access. *Directions:* wearable heat-detection for cattle; low-cost mastitis strip; backyard poultry climate shelter.

4. Water Management & Conservation

Focus: wells, laterite aquifer recharge, drinking-water quality, flood/drought, greywater reuse. *Directions:* well-recharge filter for households; fluoride/iron field kit; apartment greywater loop.

5. Renewable Energy & Energy Systems

Focus: rooftop solar O&M, micro-hydro, biomass, storage, energy access for coastal/hilly hamlets. *Directions:* auto solar-panel cleaner (dust/soot); pico-hydro for streams; biogas from fish-market waste.

6. Transport, Automotive & Harbour Technologies

Focus: last-mile, fishing-harbour logistics, EV charging, road safety, inland waterways. *Directions:* fish-landing cold-transport box; low-cost blind-spot alert for buses; e-rickshaw swappable pack.

7. Climate Change Adaptation & Environmental Science

Focus: sea-level rise, landslides (Wayanad/Idukki), heat islands, carbon accounting. *Directions:* landslide early-warning mesh; coastal-erosion monitor; school heat-action kit.

8. Solid, Liquid & Hazardous Waste Management

Focus: source segregation, Haritha Karma Sena workflows, e-waste, septage, hospital waste. *Directions:* bio-bin for flats; sanitary-pad incinerator; septage-tracking app for panchayats.

9. Modern Medicine, Public Health, Biomedical Technology

Focus: NCD screening, TB/leptospirosis, emergency referral, low-cost diagnostics, assistive care. *Directions:* anaemia triage device for camps; bedsore-prevention mattress overlay; fumigation device for wards.

10. Women, Children, Elderly, PwD & Marginalised Communities

Focus: safety, anganwadi nutrition, elder fall-risk, disability access, tribal outreach. *Directions:* fall-detection band for elderly living alone; tactile learning kit; low-cost sanitary vending + disposal for schools.

11. Future Technologies & Artificial Intelligence

Focus: Malayalam NLP, edge AI, robotics, digital twins, AI for governance/agri/health. *Directions:* Malayalam voice helpdesk for schemes; crop-disease vision model on phone; traffic-violation detection aid.

12. Traditional Industries, MSME & Market Innovation

Focus: coir, handloom, cashew, khadi, market linkage, packaging, GI products. *Directions:* coir-pith value board; handloom e-catalogue + costing tool; cashew-shell bio-fuel.

13. Education, Skills & Human Development

Focus: learning gaps, labs in Malayalam medium, skilling, career guidance, teacher tools. *Directions:* low-cost science kit for UP schools; skill-matching app for ITI; attendance + learning tracker for multi-grade rooms.

14. Culture, Arts & Creative Technologies

Focus: Theyyam/Kathakali documentation, mural conservation, creator economy, AR/VR heritage. *Directions:* AR walk for museums/forts; digitisation rig for palm-leaf manuscripts; royalty tracker for folk artists.

15. Good Governance & Public Service Delivery

Focus: grievance redressal, queue management, land records, panchayat dashboards, RTI. *Directions:* token + SMS queue for taluk offices; pothole-report + fix-closure loop; file-movement tracker.

16. Space, Aeronautics & Defence Technologies

Focus: CanSat/UAV, coastal surveillance aid, disaster mapping, sat-data for fisheries/agri. *Directions:* student CanSat telemetry kit; low-cost flood-map drone; fish-aggregation forecast display.

17. Fisheries & Ocean Science

Focus: catch preservation, ghost nets, craft safety, value addition, marine litter. *Directions:* solar ice-box for vallams; ghost-net collection incentive; onboard distress beacon.

18. Infrastructure Development & Livelihood Systems

Focus: rural roads, housing (LIFE Mission), public-building maintenance, livelihood diversification. *Directions:* concrete-maturity meter for small contractors; low-cost laterite-block tester; SHG livelihood planner.

19. Tourism & Hospitality

Focus: responsible tourism, homestays, waste at destinations, safety, multilingual guides. *Directions:* carrying-capacity counter for eco-spots; homestay hygiene rating; offline audio guide.

20. Financial Systems & FinTech

Focus: Kudumbashree credit, chit-fund transparency, fraud alerts, micro-insurance. *Directions:* SHG ledger app in Malayalam; phishing-alert tutor for seniors; micro-premium collector.

21. Supply Chain & Logistics

Focus: farm-to-mandi, cold chain breaks, last-mile in ghats, warehouse tracking. *Directions:* agri-vehicle pooling board; crate tracker; mandi-price SMS board.

22. Disaster Management & Public Safety

Focus: floods (2018/19 lessons), fire, drowning, crowd safety, first response. *Directions:* river-level SMS alarm; low-cost life-buoy drone aid; school-evacuation drill kit.

23. Sports, Wellness & Fitness

Focus: school fitness, physio access, traditional games (Kalaripayattu), nutrition. *Directions:* posture-screening app; low-cost agility kit; hydration + heat guide for athletes.

24. Biotechnology, Molecular Biology, Genetics & Life Science

Focus: diagnostics, fermentation, tissue culture (banana/orchid), water-hyacinth bio-products. *Directions:* rapid water-pathogen strip; banana tissue-culture hardening chamber; hyacinth fibre composite.

25. Complementary and Alternative Medicine – AYUSH

Focus: Ayurveda standardisation, raw-drug traceability, wellness protocols, documentation. *Directions:* herb-adulteration lens; dosage-reminder for regimens; raw-drug provenance log.

26. Earth Sciences, Mining & Geology

Focus: quarrying impacts, laterite/clay mapping, groundwater prospecting, coastal sand. *Directions:* quarry-vibration logger; well-yield estimator; landslide-susceptibility school map.

27. Assistive Technologies & Wearables

Focus: low-vision/hearing aids, mobility, prosthetics, caregiver support, classroom inclusion. *Directions:* haptic navigation band; speech-to-Malayalam-text aid; adjustable low-cost walker.

How to get the verbatim 1972 problem statements

The shelf is not public HTML – it loads as authenticated API data after login. Steps:

1. Sign up and sign in as in §2. Use desktop Chrome/Edge; disable aggressive ad-block for `yip.kerala.gov.in`.
2. Dashboard → `Idea submission` → `Choose from problem shelf`. Filter by theme (above 27), district, department.
3. Open each item to copy: Problem ID, Title, Department/Organisation, Contact, full text (≥100 words), expected outcome.
4. Paste into §7 template below, one per `\subsection`, cite ID + URL + access date for traceability.
5. If dashboard paginates or virtualises, scroll + "Load more" to the end before printing to PDF; 1972 rows typically need batched copy in chunks of 50–100.

Do not share your password or session tokens with anyone including automated scrapers; institutional labs should use their own logins.

Idea-submission template (use one copy per problem)

Problem ID: ____ **Theme:** ____ **Dept:** ____
Title: ____
Verbatim statement (paste): ____
Who is affected / where: ____
Current gaps: ____
Outcome needed: ____
Proposed approach (DUI + IR4.0): ____ **Team:** ____

Problem-statement submission (for departments)

Departments submit via https://yip.kerala.gov.in/problemstatement: Organisation name, point-of-contact, email, phone, district, title, statement (min 100 words covering challenge, affected population, gaps, outcome). Review before Submit.

Sources and verification

- Homepage + journey + stats + startups: https://yip.kerala.gov.in/ (fetched 2026-10-11).
- About, quadruple helix, committees, example projects: https://yip.kerala.gov.in/about.
- 27 theme titles verbatim: https://yip.kerala.gov.in/theme.
- Submission form fields + 100-word rule: https://yip.kerala.gov.in/problemstatement.
- Signup fields + verify-email gate: https://yip.kerala.gov.in/student/signup.
- K-DISC YIP overview + 30 areas / 1972 shelf / L0-L1 flow: K-DISC Annual Report 2025-26 https://kdisc.kerala.gov.in/wp-content/uploads/2026/05/K-DISC_2025-26_Annual-Report_240226.pdf and https://kdisc.kerala.gov.in/en/young-innovators-programme-yip.
- YIP 8.0 window 01-Oct-2025–15-Nov-2025: DTE circular https://dtekerala.gov.in/static/DEPTCIRCULARS/document-4433_merged.pdf; scheme approved 30-12-2022: https://www.dtekerala.gov.in/static/GOVTCIRCULARS/YIP.pdf.
- Crawl: UmbraCrawl `research` attempted for shelf queries (9+17 crawled, blocked by Next.js/auth); `/student` without session returns 404.